

trol, repeatability, part supply and the like. Finally, neither of these solutions address real problems.

Nynne Sole Dala, founder, mmWave Test Solutions, says, “A test chamber for testing millimetre-wave (mmWave) products, including 5G unlicensed band (57GHz to 71GHz) and 5G (5G NR, 28GHz) licensed band, is what telecom infrastructure companies are investing in, for both 5G backhaul and fixed wireless access (FWA).

“FWA replaces the expensive last 100 metres of fibre from the curb to the building. 802.11ad and 802.11ay (unlicensed 5G) delivers gigabit speeds. India is looking at expanding the use of these technologies for backhaul. From a test perspective, the 77GHz ADAS/radar test market is growing rapidly. Our MilliBox system is aimed at 20GHz - 95GHz, and does not support 4G frequencies (sub-3GHz).”

James Kimery, director of wireless research at National Instruments (NI), says, “NI recently announced two series of mmWave radio heads for mmWave transceiver system. The radio heads, which cover the spectrum from 24.5GHz to 33.4GHz and 37GHz to 43.5GHz, are targeted at wireless researchers prototyping 5G NR systems. These are compatible with the mmWave transceiver system, including base-band sub-system and software. These are also interchangeable with previously-released mmWave radio heads. This means that existing software can be reused with minimal changes.”

## Challenges and areas for development

The biggest challenge for T&M in the transition from 4G to 5G is that  $\mu$ Wave (for 4G) paradigm always look at shielding as an important feature for a chamber. Whereas, for mmWave, shielding is secondary to addressing stray reflection issues inside the chamber.

$\mu$ Wave engineers treat antennae as an independent entity. This does not work well in mmWave where the antenna array cannot be isolated easily, either from the IC's power amplifiers, phase shifters and splitters, or from beamforming algorithms, as these all contribute to achieving the desired propagation performance. This means, radiation pattern measurement is no

## Some T&M solutions for 5G

1. A broadband series system, like VectorStar's ME7838D, measures wide-band modulated S parameters for characterisation of  $\mu$ Wave and mmWave devices under modulated conditions.
2. RF drive testing and optimisation of networks can be used for benchmarking, self-optimised network performance or simply KPI analysis.
3. A radio communication test station with 5G base station emulation function can support both sub-6GHz and mmWave bands used by 5G. mmWave-band RF measurements and beamforming tests can be achieved by combining it with OTA chamber using call connections specified by 3GPP. One such instrument is Anritsu's MT8000A.
4. An arbitrary waveform generator can produce wide-band waveforms needed to test 5G systems. It can also be used to drive a vector signal generator. For example, Keysight's two-channel M8190A offers 5GHz analogue bandwidth and up to 90dBc spurious-free dynamic range.
5. A microwave analogue signal generator provides LO signals for mmWave up-converters and down-converters. One example is Keysight's precise N5183B MXG.
6. A wireless communication tester supports all cellular technologies compatible with 3GPP/3GPP2 UE RF TRX tests, including GSM/EGPRS, Cat-M1/NB-IoT, W-CDMA/HSPA, TD-SCDMA/HSPA, CDMA2000 1X/1xEV-DO and LTE/LTE-Advanced.
7. Radiation pattern measurement for 5G covers 18GHz to 95GHz with 50dB absorption. Fastest growing applications are 5G at 24GHz, 28GHz and 39GHz, and automotive radar at 77GHz and 81GHz. There is also strong activity at 60GHz for 802.11ad/ay, access and backhaul infrastructure and other proprietary unlicensed implementations.



Fig. 2: MilliBox is a test chamber for testing mmWave, by mmWave Test Solutions

longer something that is done once when the antenna is designed and forgotten. It is a continuous process, from design to production.

Testing protocols for 5G are still in the final stages of approval from 3GPP. A challenge faced while developing this drive testing tool is extraction of layer three messages from chip sets. (Every chipset has its own protocol.)

Bringing all parameters to one handheld device is complex. It depends on memory size, processor speed and more.

Cloud services and SDN/NFV introduce many new test requirements and challenges for 5G network testing.

With faster processing speeds and improved latency, new electronic circuits and hardware development are a challenge.

Low noise floor and low power consumption are other challenges.

Vikram Khurana, general manager - sales, at Microlease, says, “5G is set to offer exponential improvements

in every area compared to prior networks. These include speed, latency, bandwidth, reliability and availability. Speed and latency are especially important and what will primarily set 5G apart is enabling applications that simply could not be accomplished with prior networks.

“In terms of speed, there will be massive improvements—on the order of 10 to 100 times faster than 4G. Speeds of 10 Gigabits per second are expected. Companies like AT&T and Verizon have already achieved speeds in this range.

“In terms of latency, current 4G average in the US is 61 milliseconds. As a reference point, the blink of an eye is around 350 milliseconds, so one-sixth of that time is fast but lag or delay is still noticeable.

“We have test setups ready for signal generation and analysis, pre-5G NR solutions, antenna array testing and OTA. mmWave, small cell, massive MIMO, beamforming and full duplex are the new targeted applications.”

3GPP, the 5G standards body, officially approved the first round of 5G specifications in June 2018. This standard will be followed by the specifications of new architectural options and additional capabilities. As the standards evolve, test solutions should allow future upgrades while avoiding costly hardware changes. Commercialisation of mmWave technology is accelerating. There is a global consensus that it will be critical to achieving the goals set forth by 3GPP for 5G. **EFY**